Personalised Learning Recommendation System

Product Design Document

Online Education Platform | April 2025

1. User Stories

The following user stories capture the core needs of each persona and are directly traceable to the three business objectives: increase user engagement by 25%, improve course completion rates by 30%, and enhance user satisfaction scores by 20%.

1.1 Learners

Story-01

As a **learner**, I want the platform to recommend courses based on my declared career goals and current skill level so that I can follow a structured, relevant learning path without spending time searching manually.

Story-02

As a **learner**, I want to receive adaptive recommendations that evolve as I complete modules and assessments so that the system stays aligned with my growing proficiency and shifting interests over time.

Story-03

As a **learner**, I want to see why a course is being recommended to me — whether it's because of my career goal, a skill gap, or peer popularity — so that I can make an informed decision before enrolling.

Story-04

As a **learner**, I want to be notified when a highly relevant course is added to the catalogue or goes on offer so that I never miss an opportunity that aligns with my learning plan.

Story-05

As a **learner**, I want to set my weekly learning availability and preferred content format (video, text, project-based) so that recommendations respect my schedule and learning style, increasing the likelihood I'll actually complete them.

1.2 Instructors

Story-01

As an **instructor**, I want to see which learner segments are being recommended for my course — by skill level, career goal, and demographics — so that I can tailor my course content and positioning to better serve that audience.

Story-02

As an **instructor**, I want to receive suggestions on skill gaps that are underserved in my subject area so that I can create new courses or modules that have high demand and recommendation potential on the platform.

Story-03

As an **instructor**, I want to understand at which specific module learners who were recommended my course tend to drop off so that I can redesign those sections to improve completion rates.

Story-04

As an **instructor**, I want my newly published courses to be eligible for early recommendation trials to a relevant cohort so that I can gather engagement data quickly and iterate before a wider rollout.

Story-05

As an **instructor**, I want to tag my courses with prerequisite skills, target roles, and difficulty levels so that the recommendation engine surfaces my content to the most appropriate learners and reduces mismatched enrolments.

1.3 Platform Admins

Story-01

As a **platform admin**, I want a dashboard that shows recommendation acceptance rates, click-throughs, and downstream completion rates by algorithm variant so that I can evaluate and tune the recommendation engine with confidence.

Story-02

As a **platform admin**, I want to configure business rules — such as promoting sponsored courses or suppressing underperforming content — that layer on top of the algorithm so that commercial and quality objectives are met without fully overriding personalisation.

Story-03

As a **platform admin**, I want to run A/B tests on different recommendation strategies (collaborative filtering vs. content-based vs. hybrid) so that I can make data-driven decisions on which model best drives engagement and completion KPIs.

Story-04

As a **platform admin**, I want to receive automated alerts when recommendation diversity drops below a threshold — indicating the system is over-indexing on a narrow set of courses — so that I can intervene before learner experience degrades.

Story-05

As a **platform admin**, I want to audit the recommendation logic for bias across learner demographics (age, geography, prior education) so that the platform delivers equitable course discovery and meets compliance obligations.

2. Core Functionalities

The twelve features below define the minimum viable functional scope of the system, organised by user type and prioritised against the three business KPIs.

#	User Type	Feature	Description	Priority
1	Learner	**Onboarding Profiler**	Captures career goals, skill levels, content format preferences, and weekly availability during sign-up to seed the recommendation engine with a meaningful baseline.	High
2	Learner	**Personalised Course Feed**	Surfaces a dynamically ranked list of course recommendations on the learner's homepage, refreshed as their activity, completions, and stated preferences evolve.	High
3	Learner	**Recommendation Explainability**	Displays a plain-language rationale alongside each recommendation so learners can trust and act on suggestions.	High
4	Learner	**Adaptive Learning Path Builder**	Sequences courses into a structured path toward a career goal, automatically updating remaining steps as the learner completes milestones or skips content.	High
5	Learner	**Skill Gap Identifier**	Compares a learner's current skill set against the competency requirements of their target role and maps gaps to specific recommended courses.	Medium
6	Learner / Instructor	**Behavioural Signal Engine**	Continuously ingests implicit signals — watch-time, quiz scores, ratings, drop-off points — and feeds them back into the recommendation model to improve relevance over time.	High
7	Instructor	**Course Metadata & Taxonomy Manager**	Allows instructors to tag courses with prerequisites, difficulty, duration, target roles, and learning outcomes — the structured input the recommendation engine depends on.	High
8	Instructor	**Audience & Drop-off Analytics**	Provides instructors with a breakdown of which learner segments were recommended their course, enrolment conversions, and modules where learners disengage.	Medium
9	Platform Admin	**Recommendation Engine Control Panel**	Enables admins to configure recommendation strategies, adjust weighting parameters, and apply business rules such as promoted or suppressed content.	High
10	Platform Admin	**A/B Testing & Experimentation Framework**	Supports controlled experiments across recommendation variants with statistical significance reporting to guide model decisions.	Medium

#	User Type	Feature	Description	Priority
11	Platform Admin	**Equity & Bias Audit Dashboard**	Monitors recommendation distribution across learner demographics to detect and flag algorithmic bias and support compliance requirements.	**Medium**
12	Platform Admin	**Engagement & Performance Dashboard**	Consolidates platform-wide metrics — acceptance rate, click-through, enrolment conversions, and completion rates — to track progress against the three business KPIs.	**High**

3. Non-Functional Requirements

The following NFRs define the quality attributes the system must satisfy beyond functional correctness.

3.1 Performance

NFR ID	Name	Description
NFR-01	**Recommendation Response Latency**	The personalised course feed must load within 2 seconds at the 95th percentile under normal traffic conditions. Latency beyond this threshold directly degrades the learner's first impression and undermines the engagement KPI from the very first interaction.
NFR-02	**Real-Time Signal Processing Lag**	Behavioural signals (course completion, quiz score, video drop-off) must be ingested and reflected in the recommendation model within 5 minutes of the triggering event. A stale model risks surfacing content the learner has already consumed or explicitly skipped.

3.2 Scalability

NFR ID	Name	Description
NFR-01	**Concurrent User Throughput**	The recommendation engine must sustain performance benchmarks for up to 500,000 concurrent users, with horizontal scaling triggered automatically when CPU or memory utilisation crosses 70% on any serving node.
NFR-02	**Catalogue and Data Volume Scalability**	The system must handle a catalogue of up to 100,000 active courses and 1 billion behavioural events without degradation in quality or query performance. Model retraining pipelines must complete within a 4-hour batch window regardless of dataset size growth.

3.3 Availability

NFR ID	Name	Description
NFR-01	**Uptime SLA**	The system must maintain 99.9% uptime (no more than ~8.7 hours of unplanned downtime annually), with an RTO of 15 minutes and an RPO of 1 hour. Extended outages directly translate to learner churn and instructor revenue loss.
NFR-02	**Graceful Degradation**	In the event of a recommendation engine failure, the platform must fall back to a non-personalised popularity-based feed within 30 seconds — ensuring learners still have a functional discovery experience rather than a broken page.

3.4 Compliance

NFR ID	Name	Description
NFR-01	**Data Privacy Regulation Adherence**	The system must comply with GDPR, CCPA, and equivalent applicable regulations in all operational geographies. This includes explicit consent before collecting behavioural data, honouring right-to-erasure requests within 72 hours, and maintaining a full ROPA.
NFR-02	**Algorithmic Transparency Obligation**	Where regulations require it (e.g. the EU AI Act), the platform must produce a human-readable explanation of any individual recommendation decision on request — auditable at the model layer for regulatory purposes.

3.5 Security & Data Retention

NFR ID	Name	Description
NFR-01	**Learner Data Encryption and Access Control**	All PII and behavioural data must be encrypted at rest (AES-256) and in transit (TLS 1.3). Access is governed by RBAC with platform admins limited to aggregated, anonymised views unless explicitly authorised. Every access generates an immutable audit log entry.
NFR-02	**Behavioural Data Retention Policy**	Raw learner interaction data must be retained for a maximum of 24 months from collection, then automatically purged or irreversibly anonymised. Aggregated de-identified training datasets may be retained indefinitely. Retention schedules must be configurable by geography.

3.6 Monitoring & Logging

NFR ID	Name	Description
NFR-01	**Operational Observability**	The system must emit structured logs and real-time metrics covering latency, signal ingestion lag, inference errors, cache hit/miss ratios, and fallback activation events. Alerts must notify the on-call engineering team within 3 minutes of any threshold breach.
NFR-02	**Recommendation Quality Monitoring**	Beyond infrastructure metrics, the system must track model-level health indicators including recommendation diversity score, click-through rate by cohort, and acceptance rate drift. Automated alerts must fire if acceptance rate drops more than 15% week-over-week.

3.7 Extensibility

NFR ID	Name	Description
NFR-01	**Modular Recommendation Algorithm Architecture**	The recommendation engine must be designed as a pluggable, modular system where individual algorithm components can be swapped, layered, or replaced without requiring changes to the serving layer or downstream APIs.
NFR-02	**Third-Party Integration Readiness**	The system must expose a well-documented, versioned REST and/or GraphQL API allowing external systems — enterprise LMS platforms, HR tools like Workday or SAP SuccessFactors — to query personalised recommendations for their own users.

4. System Component Flowchart

The diagram below illustrates how the four major system layers interact to deliver personalised recommendations to learners.

LAYER 1: DATA COLLECTION & ANALYSIS

Onboarding Profiler	Behavioural Signal Engine	Course Metadata Store	Third-Party Signals
Career goals, skill level, content preferences, availability	Watch-time, quiz scores, drop-off points, ratings	Tags, prerequisites, difficulty, target roles, duration	LMS imports, HR tool data, partner content feeds

▼ Data flows into the Feature Store & Model Training Pipeline ▼

LAYER 2: RECOMMENDATION ENGINE

Collaborative Filtering	Content-Based Filtering	Hybrid Ranker + Business Rules
Learner-to-learner similarity; peer-based course surfacing	Skill gap + metadata matching; career goal alignment	Blended score + admin overlays; promoted/suppressed content

Explainability Module → Generates plain-language rationale for each recommendation

Fallback: Popularity-Based Feed → Activates within 30 seconds on engine failure

▼ Ranked recommendations served via API to the UI layer ▼

LAYER 3: USER INTERFACE

Learner Homepage Feed	Adaptive Learning Path	Instructor Analytics View	Admin Control Panel
Personalised course cards with explanation tags	Goal-based course sequence with progress tracker	Audience segments, enrolment conversion, drop-off modules	Engine config, A/B tests, bias audit, KPI dashboard

▼ Interactions feed back to Layer 1 as new behavioural signals ▼

LAYER 4: PLATFORM INTEGRATION

REST / GraphQL API	Notification Service	Observability & Logging
Versioned external API for enterprise LMS & HR tools	Push, email & in-app alerts for new relevant courses	Push, email & in-app alerts for new relevant courses

Note: *Arrows indicate primary data flow direction. The feedback loop from Layer 3 (UI) back to Layer 1 (Data Collection) is the mechanism through which the system continuously improves recommendation quality without manual retraining triggers.*

5. Major Trade-Off Analysis

The Three-Way Tension: Personalisation Depth vs. Real-Time Performance vs. Privacy & Tracking

Context

Building a personalised recommendation system forces a fundamental choice about which dimension to optimise first — because optimising all three simultaneously is architecturally and ethically impossible at launch. The three competing pressures are:

- Personalisation Depth: the richness of the model — how many signals, how much user history, how sophisticated the algorithm — directly drives recommendation relevance and completion rates.
- Real-Time Performance: the speed at which recommendations are served and updated. Deep personalisation requires compute-intensive models that conflict with the 2-second latency NFR at scale.
- Privacy & Tracking: the volume of behavioural data collected. More tracking unlocks better personalisation but increases regulatory exposure, erodes user trust, and creates long-term data liability.

5.1 Why You Cannot Fully Optimise All Three

If you prioritise...	You gain...	But you sacrifice...
Personalisation Depth	Higher relevance, better completion rates	Real-time speed (model inference takes longer) + privacy risk (needs more user data)
Real-Time Performance	Sub-2s latency, seamless UX at 500K concurrent users	Personalisation accuracy (simpler, faster models are less precise)
Privacy / Minimal Tracking	User trust, regulatory safety, reduced data liability	Personalisation quality (less signal = weaker recommendations = lower engagement)

5.2 The Product Decision

Decision: Prioritise Real-Time Performance as the non-negotiable constraint, pursue Personalisation Depth as the primary optimisation target, and treat Privacy & Tracking as a hard boundary — not a dial.

5.3 Rationale

Performance is the non-negotiable floor. The business objective to increase engagement by 25% is entirely dependent on learners actually landing on the recommendation feed and interacting with it. Research consistently shows that users abandon content feeds that take longer than 2-3 seconds to load. No matter how accurate the recommendations are, they have zero impact if the feed is too slow to render. Performance is therefore a prerequisite for the other two objectives — not a trade-off against them.

Personalisation depth is the primary lever for the KPIs. Both the 25% engagement increase and the 30% completion rate improvement are directly caused by recommendation relevance. A system that surfaces the right course to the right learner at the right moment in their path is the core value proposition. This means investing in a hybrid model (collaborative filtering + content-based filtering) from day one, not starting with a simple popularity feed and iterating. Starting shallow makes it technically harder to deepen the model later once learner expectations have been set.

Privacy is a hard boundary, not a tuning parameter. Rather than treating data collection as a lever to pull for better personalisation, the system should be designed to maximise recommendation quality within consent-first, GDPR-compliant data boundaries. This means relying on explicit onboarding signals (US-L05), transparent opt-in for behavioural tracking (NFR-C01), and anonymised aggregation for collaborative filtering — rather than maximising raw data collection. This approach protects the platform from regulatory risk, builds user trust that sustains long-term engagement, and directly supports the 20% satisfaction score improvement by ensuring learners feel in control of their data.

5.4 Implementation Consequences of This Decision

- The recommendation model uses pre-computed embeddings served from a low-latency cache rather than real-time inference — achieving the 2-second NFR without sacrificing model depth.
- Batch retraining runs every 4 hours (NFR-S02) using consented behavioural signals, keeping recommendations fresh without requiring real-time computation at the serving layer.
- The Explainability Module (NFR-C02) is built in from day one — not bolted on later — because transparency is both a regulatory requirement and a trust mechanism that improves recommendation acceptance rates.
- The A/B Testing Framework (US-A03) allows the team to progressively push toward deeper personalisation models as the dataset matures, without changing the serving architecture.

Summary

The right call for this platform — given its specific KPIs — is not to find a compromise between all three dimensions, but to establish a clear hierarchy: Performance as the floor, Personalisation as the ceiling to keep raising, and Privacy as the walls that define the space in which the system can operate. A platform that is fast, increasingly smart, and trustworthy will consistently outperform one that sacrifices any of these properties chasing short-term metric gains.